

4. Prove by induction that for $n \in \mathbb{N}$

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & -2n \\ 0 & 1 \end{pmatrix}$$

(5)

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6. Prove by induction that for all positive integers n

$$f(n) = 3^{2n+4} - 2^{2n}$$

is divisible by 5

(6)

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6. Prove by induction that, for all positive integers n ,

$$\sum_{r=1}^n (2r-1)^2 = \frac{1}{3}n(4n^2-1) \quad (6)$$



6. (i) Prove by induction that for $n \in \mathbb{Z}^+$

$$\sum_{r=1}^n (3r+1)(r+2) = n(n+2)(n+3) \quad (6)$$

- (ii) Prove by induction that for all positive **odd** integers n

$$f(n) = 4^n + 5^n + 6^n$$

is divisible by 15

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6. Given that

$$y = e^{2x} \sinh x$$

prove by induction that for $n \in \mathbb{N}$

$$\frac{d^n y}{dx^n} = e^{2x} \left(\frac{3^n + 1}{2} \sinh x + \frac{3^n - 1}{2} \cosh x \right)$$

(6)

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